

# Taehoon Kim

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## Research Interests

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My research aims to develop robust perception for real-world, safety-critical environments through multimodal representation learning with computer vision and vision–language models. I am particularly interested in spatial intelligence and autonomous-driving perception via multi-sensor fusion (RGB, LiDAR, Radar). I also study industrial and multimodal anomaly detection via distillation-based teacher–student frameworks.

**Keywords:** Computer Vision, Spatial Intelligence, Multi-Sensor Fusion, Anomaly Detection, Multimodal

## Education

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### Kyonggi University

M.S. in Computer Science

- Advisor: Junho Ahn
- GPA: 4.5 / 4.5

Sep. 2025 – Present

*Suwon, Korea*

### Kyonggi University

B.S. in Department of Restaurant and Culinary

B.S. in Department of Artificial Intelligence – *double major*

- CGPA: 4.3 / 4.5 – Summa Cum Laude
- AI GPA: 4.3 / 4.5 (42 credits completed)
- Valedictorian, Restaurant and Culinary

Mar. 2019 – Aug. 2025

*Seoul, Korea*

*Suwon, Korea*

## Experiences

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### K-SW Square, Purdue University

Research Intern

*Advisor: Eric T. Matson & Anthony Smith*

Oct 2023 – Dec 2023

*West Lafayette, IN, USA*

### AI Vision Lab, Kyonggi University

Undergraduate Research Assistant

*Advisor: Prof. Junho Ahn*

May 2024 – Aug 2025

*Suwon, Korean*

### Military–Academia Joint AI Education Program, ROK DCC

Teaching Assistant & Project Mentor

2024 – 2025

*Suwon, Korean*

### National Center of Excellence in Software, Kyonggi University

Graduate Teaching Assistant

Sep. 2025 – Present

*Suwon, Korean*

## Publications

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### International Journal

- [1] **Taehoon Kim**, Sehun Lee, Namgi Kim, Youjin Seo, “XSANet: Industrial X-ray Anomaly Detection via Spatial Transformer and Attention-Enhanced Teacher-Student Framework,” *Image and Vision Computing (IMAVIS)*, Elsevier, *under review*
- [2] **Taehoon Kim**, Sehun Lee, Junho Ahn, “Zero-Shot Vision-Based Robust 3D Map Reconstruction and Obstacle Detection in Geometry-Deficient Room-Scale Environments,” *Computers, Materials & Continua*, vol. 86, no. 2, pp. 1–30, Dec. 2025.
- [3] Sehun Lee, **Taehoon Kim**, Sookyun Kim, Junho Ahn, Namgi Kim, “Unsupervised Structural Defect Classification via Real-Time and Noise-Robust Method in Smartphone Small Modules,” *Electronics (MDPI)*, vol. 14, no. 17, art. no. 3455, Aug. 2025.
- [4] Sehun Lee, **Taehoon Kim**, Junho Ahn, “Zero-Shot Based Spatial AI Algorithm for Up-to-Date 3D Vision Map Generations in Highly Complex Indoor Environments,” *Computers, Materials & Continua*, vol. 84, no. 2, pp. 3623–3648, Jul. 2025.

### International Conference

- [1] Doyoung Koh, **Taehoon Kim**, Sehun Lee, Namgi Kim, “Weighted Knowledge Distillation with DisturbLabel for Robust Learning,” 2026 17th International Conference on Ubiquitous and Future Networks (ICUFN), Milan, Italy, Jul. 2026.

## Domestic Journal

- [1] **Taehoon Kim**, Jiyoung Na, Jiwon Yoon, Sehun Lee, Junho Ahn, “Multiple Camera-Based Real-Time Long Queue Vision Algorithm for Public Safety and Efficiency,” Journal of the Korea Society of Computer and Information (JKSCI), vol. 29, no. 10, pp. 47–57, Oct. 2024. DOI: 10.9708/jksci.2024.29.10.047

## Domestic Conference

- [1] **Taehoon Kim**, Sehun Lee, Junho Ahn, “Industrial X-ray UniNet: A Robust Spatial Transformation Architecture for X-ray Anomaly Detection in Smartphone Actuator Modules,” Proceedings of the 2025 KAD Winter Conference (**Outstanding Paper Award**)
- [2] Sehun Lee, **Taehoon Kim**, Junho Ahn, “Genetic Neural Networks: Efficient Training Initialization Strategy via Weight Recombination,” Proceedings of the 2025 KAD Winter Conference (**Best Paper Award**)
- [3] **Taehoon Kim**, Sehun Lee, SeungBeom Lee, Junho Ahn, “0-VISTA: Zero-Shot Spatial AI for 3D Mapping and Detection in Complex Indoor Environments,” Proceedings of the 2025 KAD Spring Conference
- [4] Sehun Lee, **Taehoon Kim**, Doyeon Kim, Junho Ahn, “Latent Hotspot Crossover for Disentangled and Interpretable Representation Learning in Variational Autoencoders,” Proceedings of the 2025 KAD Spring Conference (**Outstanding Paper Award**)
- [5] Doyeon Kim, SeungBeom Lee, **Taehoon Kim**, Junho Ahn, “Detection of Movable Object Layout Changes in Shared-Use Facilities,” Proceedings of the 2025 KAD Spring Conference
- [6] **Taehoon Kim**, Jiyong Na, Sehun Lee, Jiwon Yoon, Junho Ahn, “Deep Learning Vision-Based Line Recognition Algorithm Suggestion for Real-Time Intercity Bus Passenger Queue Efficiency,” Proceedings of the 2024 KSII Spring Conference
- [7] Yechan Lee, **Taehoon Kim**, Jiyoung Na, Jiwon Yoon, Sehun Lee, Hanyong Lee, “Analysis of Autonomous Driving Algorithms for Road Driving of Delivery Robots,” Proceedings of the 2023 KIIT Summer Conference

## Patent

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| <b>Deep Learning-Based Queue Recognition System</b> | Oct. 2024           |
| Korea Patent Application                            | No. 10-2024-0144283 |

## Projects

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| <b>Pedestrian Safety System for Large-Scale Buildings Using 3D Congestion Mapping with Collaborative Robots and Vision AI for Disaster Response</b><br>NRF | Sep. 2025 – Aug. 2026 |
| <b>Acoustic-Based Real-Time Drone Localization &amp; Tracking AI System</b><br>ITSA & ROK DCC                                                              | Jul. 2025 – Dec. 2025 |
| <b>Improvement of Detect Detection and Classification Capabilities in SUB2 ASS'Y GAP Inspection</b><br>im co., ltd.                                        | Nov. 2024 – Feb. 2025 |
| <b>Real-Time Multimodal Vision-Integrated Military Fitness Assessment</b><br>ITSA & ROK DCC                                                                | Sep. 2024 – Dec. 2024 |
| <b>Improvement of Detect Detection and Classification Capabilities in X-RAY Ball Inspection</b><br>im co., ltd                                             | Jul. 2024 – Aug. 2024 |
| <b>Enhancing Culinary Experience: Real-Time Interactive Cooking Assistance Utilizing Mixed Reality</b><br>Purdue University                                | Oct. 2023 – Dec. 2023 |

## Awards

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| <b>Best Paper Award</b> , 2025 Korea Data Science Winter Conference | Dec. 2025 |
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| <b>Outstanding Paper Award</b> , 2025 Korea Data Science Winter Conference             | Dec. 2025 |
| <b>Outstanding Paper Award</b> , 2025 Korea Data Science Spring Conference             | Jun. 2025 |
| <b>3rd Prize</b> , Kyonggi University SW Imagination Company Project Competition       | Nov. 2024 |
| <b>1st Prize</b> , Kyonggi University Advanced Capstone Design Project Competition     | Jun. 2024 |
| <b>2nd Prize</b> , Kyonggi University Foundational Capstone Design Project Competition | Jun. 2023 |
| <b>1st Prize</b> , Kyonggi University GamsungSW IT Competency Competition              | Dec. 2019 |

## Honors

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| <b>Letter of Appreciation</b> , ROK Defense Communication Command<br>· Award for outstanding contribution in AI development and mentoring | Apr. 2026                         |
| <b>AI &amp; SW Talent Scholarship</b> , Noh Youngbaek Scholarship Foundation<br>· Award for excellence in the AI & Software field         | Nov. 2025                         |
| <b>Faculty Council Honors Scholarship</b> , Kyonggi University<br>· Award for outstanding academic achievement                            | Jun. 2025                         |
| <b>Highest Honors</b> , Kyonggi University<br>· Graduate 1st in Dept. of Restaurant & Culinary                                            | 2025                              |
| <b>Semester High Honors</b> , Kyonggi University<br>· Award for perfect GPA of 4.5 / 4.5                                                  | 2024                              |
| <b>Semester High Honors</b> , Kyonggi University<br>· Award for Academic Excellence (1st)                                                 | 2019, 2022 · 2023 (1st Sem), 2024 |

## Technical Skills

### Programming languages

Python, C, Java, R

### Deep Learning Framework

Pytorch, Tensorflow, Keras

## Language Skills

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| <b>TOEIC – Total Score 920</b>                                                                                                                                                                     | Aug. 2024                                 |
| <b>TOEIC Writing – Advanced Mid (AM)</b>                                                                                                                                                           | Apr. 2024                                 |
| <b>TOEIC Speaking – Intermediate High (IH)</b>                                                                                                                                                     | Apr. 2024                                 |
| <b>English Presentation</b><br>Kyonggi University National Center of Excellence in Software<br>· Presented program highlights and achievements to a CHED delegation, Philippine President's office | Jul. 2024<br><a href="#">news article</a> |
| <b>English Translation</b><br>Purdue Military Research Institution (PMRI) & IITP<br>· Translate AI lectures and supported communication for ROK ROTC officers                                      | Oct. 2023                                 |